Hackathon Preparation Guide

NER-SMARTLOG: AI Street Logistics & Hazard Intelligence Portal

Smart India Hackathon 2026 | Problem Statement 26002 | Ministry of Development of NER [MDoNER]

1. EXECUTIVE PROJECT SUMMARY

NER-SMARTLOG is an AI-powered predictive logistics and road hazard telemetry portal engineered specifically
for the 8 North Eastern states (Assam, Arunachal Pradesh, Meghalaya, Manipur, Mizoram, Nagaland, Tripura, and) Tj T* (Sikkim). It provides automated risk a
synchronized field-to-headquarters disaster dispatch to prevent critical supply-chain paralysis (vaccines, oxygen,) Tj T* (relief grain) caused by monsoonal land

Core System Specifications:

* Problem Statement ID : SIH 26002 [Ministry of Development of North Eastern Region - MDoNER]
* Strategic Corridors : NH-10 [Sikkim], NH-13 [Arunachal Sela Corridor], NH-29 [Nagaland], NH-37 & NH-6 [Assam/Meghalaya]

* Core Technology Stack : Python Flask, Jinja ChoiceLoader, OSRM Graph Engine, Open-Meteo NWP API, Leaflet.js GIS

2. MATHEMATICAL HAZARD RISK MODEL & SCORING EQUATION

The AI decision engine evaluates corridor segments using a weighted multi-factor vulnerability equation:

Risk Score = (0.30 x R) + (0.25 x L) + (0.20 x S) + (0.10 x T) + (0.10 x H) + (0.05 x E)

Where: R = Precipitation [mm/hr] | L = Landslide & Geological Slope Gradient | S = Ground Passability [BRO Alert]
T = Traffic Density | H = Historical Hazard Frequency | E = Altitude & Elevation Fog Index
Thresholds: 0-30: LOW [Clear] | 31-60: MEDIUM [Single-lane] | 61-80: HIGH [Caution] | 81-100: CRITICAL [Auto-Reroute]

3. TECHNICAL ARCHITECTURAL HIGHLIGHTS

* ChoiceLoader Template Resolution: Prevents TemplateNotFound by searching templates/, base dir, and working root.
* Embedded NER Gazetteer: 33 pre-indexed Himalayan transit nodes with automatic fallback to Nominatim Geocoder.
* Hybrid Routing Pipeline: Primary OSRM driving graph with geometric spline fallback ensuring 100% path generation.
* Live NWP Meteorological Telemetry: Direct REST integration with Open-Meteo models polling all 8 state capital hubs.
* NER-SMARTBOT Natural Language Engine: Dynamic intent extraction answering routes, convoy speed, and alerts in <80ms.

4. HACKATHON EVALUATION & VIVA QUESTIONS (PART 1)

CATEGORY A: DOMAIN, BASIC CONTEXT & USP

Q1. What makes NER-SMARTLOG fundamentally innovative compared to Google Maps?

Ans: Commercial navigation apps optimize strictly for shortest driving time based on current traffic speed. In the Himalayas, roads show 'clear' until a sudden landslide strikes. NER-SMARTLOG is predictive: it evaluates precipitation rates, soil saturation, and verified BRO alerts to steer relief convoys onto safer mountain bypasses before roads get choked.

Q2. What are the three command roles and their distinct access permissions?

CATEGORY B: ALGORITHMIC & TECHNICAL ARCHITECTURE

Q3. How does the AI routing engine compute and select the safest alternative corridor?

Ans: The system queries OSRM for candidate driving corridors, applies the Multi-Factor Risk Equation to each polyline, and dynamically scores them. If the primary highway exceeds Risk > 60 [e.g. NH-13 slide], the engine promotes the safest mountain bypass [e.g. Sela Tunnel Bypass] as the 'AI Optimal Corridor' with green telemetry highlight.

Q4. How does Jinja ChoiceLoader eliminate TemplateNotFound errors?

Ans: By binding ChoiceLoader([FileSystemLoader('templates/'), FileSystemLoader(BASE_DIR), FileSystemLoader('.')]), Flask reliably discovers templates whether placed in a subfolder or root, regardless of the working execution path.

CATEGORY C: HARDWARE, CONNECTIVITY & OFFLINE ARCHITECTURE

Q5. How does the platform function in zero-connectivity mountain blackspots?

CATEGORY D: WORKING, VERIFICATION & SECURITY

Q6. How does the system prevent fake disaster reports from civilian users?

Ans: Role-Based Cryptographic Access Control [RBAC]. Only verified personnel with authenticated Department Badge IDs [BRO Taskforce, NDRF, SDMA] hold broadcast permissions, ensuring verified, noise-free incident feeds.

5. HACKATHON EVALUATION & VIVA QUESTIONS (PART 2)

CATEGORY E: LIMITATIONS & EDGE CASES

Q7. What are the current limitations of the prototype?

CATEGORY F: DIFFICULT JUDGE QUESTIONS & COUNTER-DEFENSES

Q8. Judge: 'What if OSRM server goes offline or encounters a routing timeout?'

Ans: We built an integrated Geometric Spline Fallback Generator in app.py. If OSRM fails to respond within 3 seconds, the engine generates trigonometric spline corridors between geocoded coordinates, guaranteeing 100% uptime for drivers.

CATEGORY G: SUSTAINABILITY, COST & FUTURE SCOPE

Q9. How is the project scalable and sustainable for state governments?

Ans: Built 100% on open-source technologies [Python, Leaflet, OpenStreetMap, Open-Meteo], eliminating recurring GIS licenses.
Future Roadmap: Phase 2: Direct API tie-in with ISRO Bhuvan/MOSDAC landslide susceptibility satellite maps;
Phase 3: Hardwired OBD-II IoT transponders with satellite SOS beacons across state health & FCI fleets.

6. QUICK-REVISION ELEVATOR PITCHES

30-Second Elevator Pitch:

"NER-SMARTLOG is an AI street logistics and hazard intelligence portal built for MDoNER [SIH 26002]. Unlike reactive GPS apps, our platform combines live meteorological sensor feeds, geological risk scoring, and verified ground BRO telemetry to predict landslides and re-route essential convoys across 8 North-Eastern states before they get stranded."

1-Minute Comprehensive Pitch:

"Respected Judges, in North East India, a single monsoon landslide cuts off life-saving oxygen, vaccines, and food supplies to remote communities for days. Current navigation tools only show red traffic lines after a corridor is already blocked. NER-SMARTLOG solves this by introducing a predictive multi-tier intelligence grid. Our AI router evaluates live rainfall volume, slope gradient, and BRO ground reports using a weighted risk index to compute hazard-resilient bypass corridors. Featuring dedicated dashboards for BRO Field Patrols, Logistics Commanders, and SDMA Disaster Cells, NER-SMARTLOG transforms fragile mountain roads into resilient, life-saving supply lifelines."